

Total No. of Questions : 10]

SEAT No. :

P2365

[Total No. of Pages : 2

[5870] - 1533

T.E. (Computer Engineering)

EMBEDDED SYSTEMS & INTERNET OF THINGS (ES and IOT)
(2015 Pattern) (Semester - II) (310252)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates :

- 1) Answer any five questions Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8, Q.9 or Q.10.
- 2) Assume Suitable data wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Draw neat & labelled diagram wherever necessary.

Q1) a) What are the different features that an embedded system should possess?
[5]

b) Compare REST-based communication and WebSocket communication API.
[5]

OR

Q2) a) What is real time system? Explain the different real time scheduling algorithm.
[6]

b) Explain Bluetooth, one of the pillar of IoT.
[4]

Q3) a) What is SCADA? What are the different blocks of SCADA.
[5]

b) Explain the steps for Purpose & Requirements specification in IoT design methodology.
[5]

OR

Q4) a) With the help of appropriate diagram explain WebSocket-based communication APIs.
[3]

b) Draw and Explain the Four pillars of IOT.
[4]

c) What is Raspberry Pi? Explain the features of it.
[3]

P.T.O.

- Q5) a)** What are the different topology of 802.15.4? Explain with suitable diagram. [6]
b) What is BACnet? Explain the different layers function. [6]
c) Explain the Sensor Standards for Interoperability. [4]

OR

- Q6) a)** Explain the Zigbee architecture with suitable diagram. [6]
b) Discuss the issues with IoT Standardization. [4]
c) What is KNX? Explain KNX-TP features, its Telegram. [6]

- Q7) a)** What is Web of Things (WOT)? What are the two pillars of the web? [6]
b) Explain unified multitier WOT Architecture in details. [6]
c) What is OSGi: The Universal Middleware? [5]

OR

- Q8) a)** Explain Cloud Middleware Architecture. [6]
b) Explain RFID middleware standards? [6]
c) What are the different attacks are possible in IoT? [5]

- Q9) a)** Design Weather Monitoring system, what are the different components required? Draw deployment design for this system. [6]
b) Write short note on : [6]
i) Amazon EC2 ii) Xively Cloud for IoT
c) Explain Django python web application framework. [5]

OR

- Q10) a)** Explain WAMP and its key concepts with diagram. [5]
b) Explain in brief Model, Template and View in Django architecture. [6]
c) Design Air Pollution Monitoring(APM) based on followings [6]
i) Define process specification for APM IoT system
ii) Domain model of APM IoT system
iii) Information model of APM IoT system
iv) Controller service of APM IoT system

